EX:6 WRITE THE PROGRAM FOR VACUUM CLEANER AGENT

AIM

To develop a Python program that simulates a **Vacuum Cleaner Agent** to clean all dirty rooms and display the final cleaned state of the environmen

ALGORITHM

Step 1: Start the program.

Step 2: Accept the state of Room A and Room B (Clean/Dirty) from the user.

Step 3: Place the vacuum cleaner in Room A and check its state.

Step 4: If Room A is dirty, clean it; otherwise, move to Room B.

Step 5: Check the state of Room B. If it is dirty, clean it.

Step 6: Display the final state of both rooms and end the program.

PROGRAM CODE:

```
room = {
    'A': input("Enter status of Room A (Dirty/Clean): ").capitalize(),
    'B': input("Enter status of Room B (Dirty/Clean): ").capitalize()
}

location = input("Enter Vacuum Location (A/B): ").upper()

print("\nInitial Room Status:", room)

if room[location] == "Dirty":
    print("Cleaning Room", location)
    room[location] = "Clean"
else:
```

```
    print("Room", location, "is already Clean")

location = 'B' if location == 'A' else 'A'

if room[location] == "Dirty":
    print("Cleaning Room", location)
    room[location] = "Clean"
else:
    print("Room", location, "is already Clean")

print("\nFinal Room Status:", room)
```

OUTPUT:

```
====== RESTART: C:/Users/Hp/AppData/Local/Programs/Python/Python314/EX
Enter status of Room A (Dirty/Clean): dirty
Enter status of Room B (Dirty/Clean): clean
Enter Vacuum Location (A/B): A

Initial Room Status: {'A': 'Dirty', 'B': 'Clean'}
Cleaning Room A
Room B is already Clean

Final Room Status: {'A': 'Clean', 'B': 'Clean'}
>>>
```

RESULT:

The Python program successfully simulates the Vacuum Cleaner Agent by cleaning all dirty rooms.The final state of the environment is displayed, showing that all rooms are clean.